ArchTree-LLM — Overall Project Workflow

Trie-based deduplication + zero-cost proxies + hardware-aware coverage, with LLM-guided warm-start evaluated as a controlled study

TECHNOLOGY USED

Benchmarks & Data
– NAS-Bench-201 (15,625 archs)
– HW-NAS-Bench (6 devices)
– NAS-Bench-101 · NATS-Bench SSS

Language & Config
– Python
– config.yaml (single source of truth)

Search Engine
– Compressed trie (O(L) dedup)
– Rank-UCB (C = 0.3) + mutation

Zero-Cost Proxies
– SynFlow + NASWOT + SVD entropy
– label-free (0 labels); top-35% pass

Baselines
– Random · RE · Local search
– ENAS / MCTS / ZiCo / RZ-style
– LLM-NAS

Statistics
– Friedman + Nemenyi CD · Holm
– TOST · Cohen's d · P(improvement)

Storage
– JSON benchmark caches
– CSV result logs

Visualisation & UI
– Python backend + Plotly dashboard
– plot_results.py (paper figures)

WORK FLOW DIAGRAM

